

The expected characteristic polynomial of a random lift of a
cycle:
an elementary proof, a generating function,
and a machine-checked formalization

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Abstract

Let $\Phi_{n,r}(x)$ denote the expected characteristic polynomial of a uniformly random permutation r -lift of the cycle C_n . We give a short, self-contained derivation of the closed form $\Phi_{n,r}(x) = \chi_{C_n}(x) \cdot U_{r-1}(T_n(x/2))$, where T, U are Chebyshev polynomials and $\chi_{C_n}(x) = 2T_n(x/2) - 2$; equivalently, the d -matching polynomial of C_n (Hall–Puder–Sawin) is $U_d(T_n(x/2))$. This statement is equivalent, via the Hall–Puder–Sawin identity, to a conjecture of Hall proved combinatorially by Cochran, Groothuis, Herring, Rohatgi and Stucky (2018); our proof is independent, passing through the exponential formula and a Chebyshev generating function, and yields as byproducts the generating function $\sum_{d \geq 0} \mu_{d,C_n}(x) z^d = (1 - 2T_n(x/2)z + z^2)^{-1}$, a Fibonacci–Lucas evaluation, and a parity theorem for the quotient. The entire derivation from cycle weights to the closed form — including the exponential formula for S_r — is formalized in Lean 4 over Mathlib, with no axioms beyond Lean’s standard three. The same argument applies to every connected graph of first Betti number one, where it gives $\mu_{d,G} = V_d$ for the three-term recurrence $V_d = \mu_G V_{d-1} - \mu_{G-V(C)}^2 V_{d-2}$ in the ordinary matching polynomials of G and of the forest obtained by deleting the cycle; this answers Question 6.5 of Hall–Puder–Sawin for that family, and we show the Betti-number-one hypothesis cannot be relaxed.

1 Introduction and statement

The expected characteristic polynomial of a random lift is the object at the heart of the interlacing-families method of Marcus–Spielman–Srivastava [7], who used the $r = 2$ (signing) case to prove that bipartite Ramanujan graphs exist in every degree; Hall–Puder–Sawin [2] extended the framework to arbitrary coverings. We evaluate it in closed form for the cycle.

A *permutation r -lift* of a graph G replaces each vertex by r copies and each edge $\{u, v\}$ by the perfect matching $u_i \sim v_{\sigma(i)}$ of a permutation $\sigma \in S_r$ chosen independently and uniformly per edge. For the cycle C_n , gauge-fixing a spanning path reduces the lift to a single permutation $\sigma \in S_r$ on the closing edge, and the lift decomposes into disjoint cycles: each ℓ -cycle of σ contributes a cycle $C_{n\ell}$ (fixed points contribute copies of C_n). Since $\chi_{C_m}(x) =$

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$\det(xI - A_{C_m}) = 2T_m(x/2) - 2$, the expected characteristic polynomial is the expected cycle weight

$$\Phi_{n,r}(x) = \mathbb{E}_{\sigma \in S_r} \prod_{\ell \in \text{cycles}(\sigma)} (2T_{n\ell}(x/2) - 2). \quad (1)$$

Theorem 1. *For all $n \geq 1$, $r \geq 1$, with $Y = T_n(x/2)$,*

$$\Phi_{n,r}(x) = U_r(Y) - 2U_{r-1}(Y) + U_{r-2}(Y) = \underbrace{(2T_n(x/2) - 2)}_{\chi_{C_n}(x)} \cdot \underbrace{U_{r-1}(T_n(x/2))}_{\Psi_r(x)}.$$

The factored form identifies the quotient Ψ_r — by the theorem of Hall–Puder–Sawin [2] equating the expected new-eigenvalue characteristic polynomial with the d -matching polynomial ($d = r - 1$) — as

$$\mu_{d,C_n}(x) = U_d(T_n(x/2)),$$

which is equivalent (through $\mathcal{P}_d(2t) = U_d(t)$, $\mathcal{C}_n(2t) = 2T_n(t)$ and the composition identity $U_{dn+n-1} = U_d(T_n)U_{n-1}$) to *Hall's conjecture* $\mathcal{C}_{n,d} = \mathcal{P}_d(\mathcal{C}_n)$, proved in [3]. Our route is different and elementary; we make no use of matching-polynomial combinatorics.

Corollary 2. $\sum_{d \geq 0} \mu_{d,C_n}(x) z^d = \frac{1}{1 - 2T_n(x/2)z + z^2}, \quad \text{and} \quad \sum_{r \geq 0} \Phi_{n,r}(x) z^r = \frac{(1 - z)^2}{1 - 2T_n(x/2)z + z^2}.$

Corollary 3 (real-rootedness). *All roots of $\Phi_{n,r}$ lie in $[-2, 2]$: with $x = 2 \cos \theta$, χ_{C_n} vanishes at $\theta = 2\pi m/n$ and $\Psi_r = \sin(rn\theta)/\sin(n\theta)$ vanishes at $\theta = m\pi/(rn)$ with $r \nmid m$ (when $r \mid m$ the denominator vanishes too and $\Psi_r = \pm r \neq 0$); this leaves exactly $n(r - 1)$ roots, matching $\deg \Psi_r$.*

Corollary 4 (Fibonacci–Lucas evaluation). $\Phi_{n,r}(3) = (L_{2n} - 2) F_{2nr}/F_{2n}$, since $T_n(3/2) = L_{2n}/2$ and $U_{r-1}(L_{2n}/2) = F_{2nr}/F_{2n}$.

Corollary 5 (parity). $\Psi_r(-x) = (-1)^{n(r-1)}\Psi_r(x)$: the quotient is supported on exponents of a single parity, like a matching polynomial. (For bipartite C_n it is an even polynomial for every r ; at $r = 2$ it is the classical matching polynomial $2T_n(x/2)$ of C_n , consistent with *Godsil–Gutman* [1].)

2 Proof

Write $f(\ell) = 2T_\ell(Y) - 2$ with $Y = T_n(x/2)$; by the composition identity $T_{n\ell} = T_\ell \circ T_n$ this is the per-cycle factor in (1).

Step 1 (exponential formula). For any commutative ring R and $f : \mathbb{N} \rightarrow R$, the S_r -average $\Phi_r = \mathbb{E}_\sigma \prod_{\ell \in \text{cycles}(\sigma)} f(\ell)$ satisfies

$$r \Phi_r = \sum_{k=1}^r f(k) \Phi_{r-k}, \quad \Phi_0 = 1, \quad (2)$$

the coefficient form of $\sum_r \Phi_r z^r = \exp(\sum_\ell f(\ell) z^\ell / \ell)$. (Proof: group σ by cycle type λ with Cauchy's count $r!/z_\lambda$, mark a part, and re-index; see the Lean file `Paper2ExpFormula.lean` for the fully detailed version.)

Step 2 (Chebyshev generating function). With $f(\ell) = 2T_\ell(Y) - 2$,

$$\sum_{\ell \geq 1} f(\ell) \frac{z^\ell}{\ell} = -\log(1 - 2Yz + z^2) + 2\log(1 - z) \implies \sum_{r \geq 0} \Phi_r z^r = \frac{(1 - z)^2}{1 - 2Yz + z^2}.$$

Extracting coefficients against $\sum U_r(Y) z^r = (1 - 2Yz + z^2)^{-1}$ gives $\Phi_r = U_r - 2U_{r-1} + U_{r-2}$, and the three-term recurrence $U_r + U_{r-2} = 2YU_{r-1}$ collapses this to $(2Y - 2)U_{r-1}(Y)$. $\square$

In the formalization the analytic Step 2 is replaced by an equivalent finite argument: the closed form is shown to satisfy (2) directly (a second-difference induction), and (2) with $\Phi_0 = 1$ determines Φ over a characteristic-zero domain.

3 Graphs of first Betti number one

The proof above uses cycles in exactly one place: gauge-fixing a spanning path leaves a *single* permutation, because C_n has $|E| - |V| + 1 = 1$. The argument therefore applies verbatim to any connected G with $b_1(G) = |E| - |V| + 1 = 1$, a cycle with trees attached, and we record what it gives.

By [2] the expected characteristic polynomial of a random r -lift of *any* graph factors as $\chi_G \cdot \mu_{r-1,G}$; the content below is therefore not that factorization but a closed form for the factor $\mu_{d,G}$ itself. Question 6.5 of [2] asks which parts of the theory of the matching polynomial $\mu_{1,G}$ generalize to $\mu_{d,G}$; the following answers it for $b_1(G) = 1$.

Let C be the unique cycle of G and let $G - V(C)$ be the forest obtained by deleting its vertices. Write μ_H for the ordinary matching polynomial of a graph H , with $\mu_\emptyset = 1$. Define

$$V_0 = 1, \quad V_1 = \mu_G, \quad V_d = \mu_G V_{d-1} - \mu_{G-V(C)}^2 V_{d-2}. \quad (3)$$

Theorem 6. *For every connected G with $b_1(G) = 1$ and every $d \geq 0$, $\mu_{d,G} = V_d$.*

For $G = C_n$ the deleted forest is empty, so $\mu_{G-V(C)} = 1$ and (3) is the recurrence for $U_d(\mu_{C_n}/2) = U_d(T_n(x/2))$: Theorem 1 is the case $\mu_{G-V(C)} = 1$. In general the attached trees enter only through $\mu_{G-V(C)}$; for the triangle with one pendant edge, $\mu_G = x^4 - 4x^2 + 1$ and $\mu_{G-V(C)} = x$.

The proof goes through the twisted characteristic polynomial. Let $e = uv$ be the unique non-tree edge, let $A_G(z)$ be the adjacency matrix of G with the (u, v) entry replaced by z and the (v, u) entry by z^{-1} , and set $F(x, z) = \det(xI - A_G(z))$.

Lemma 7. $F(x, z) = \mu_G(x) - \mu_{G-V(C)}(x)(z + z^{-1})$.

Proof. By the Sachs coefficient theorem $\det(xI - A)$ expands over elementary subgraphs, that is disjoint unions of edges and cycles. Since a permutation uses each of the two twisted entries at most once and $A_G(z)^\top = A_G(z^{-1})$, the determinant is of the form $P(x) + Q(x)(z + z^{-1})$. Replacing z by ± 1 and averaging kills exactly the elementary subgraphs whose cycle traverses e ; as G has the single cycle C , those are the subgraphs containing C , so $P = \mu_G$, and the surviving terms contribute $Q = -\mu_{G-V(C)}$, a cycle component carrying the Sachs weight -2 . $\square$

Proof of Theorem 6. Gauge-fix a spanning tree: the r -lift is determined by one permutation $\sigma \in S_r$ on e , and its components are indexed by the cycles of σ , the component of an ℓ -cycle being the ℓ -fold cyclic cover G_ℓ . Decomposing the $\mathbb{Z}/\ell$ action into characters gives the classical factorization $\chi_{G_\ell}(x) = \prod_{j=0}^{\ell-1} F(x, \omega^j)$ with $\omega = e^{2\pi i/\ell}$ [5, 6]. With P, Q as in Lemma 7 and $y = -P/(2Q)$,

$$\chi_{G_\ell} = \prod_{j=0}^{\ell-1} \left(P + 2Q \cos \frac{2\pi j}{\ell} \right) = (-Q)^\ell (2T_\ell(y) - 2),$$

so by the exponential formula (2), with $w = -Qz$,

$$\sum_{r \geq 0} \Phi_{G,r} z^r = \exp \left(\sum_{\ell \geq 1} (2T_\ell(y) - 2) \frac{w^\ell}{\ell} \right) = \frac{(1-w)^2}{1-2yw+w^2},$$

and extracting coefficients as in Section 2 gives $\Phi_{G,r} = \chi_G \cdot (-Q)^{r-1} U_{r-1}(y)$. Comparing with $\Phi_{G,r} = \chi_G \cdot \mu_{r-1,G}$ [2] yields $\mu_{d,G} = (-Q)^d U_d(y)$, which satisfies (3). $\square$

Since $U_d(y) = 2^d \prod_{k=1}^d (y - \cos \frac{k\pi}{d+1})$, the theorem can be restated as a product, $\mu_{d,G}(x) = \prod_{k=1}^d F(x, e^{ik\pi/(d+1)})$, whence:

Corollary 8. *For $b_1(G) = 1$, $\mu_{d,G}$ is a product of d real-rooted polynomials, and its roots are exactly the eigenvalues of the Hermitian matrices $A_G(e^{ik\pi/(d+1)})$, $1 \leq k \leq d$.*

This is the real-rootedness theorem of [2] for this family, with the roots identified.

The roots can be located much more precisely than that. Since $b_1(G) = 1$ the group $\pi_1(G)$ is $\mathbb{Z}$, so the universal cover T of G is the $\mathbb{Z}$ -cover, and T is a periodic decorated chain whose adjacency operator decomposes by Floquet–Bloch into the fibres $A_G(e^{i\theta})$. Hence

$$\text{spec}(T) = \bigcup_{\theta} \text{spec } A_G(e^{i\theta}) = \left\{ x : \exists s \in [-1, 1], \mu_G(x) = 2s \mu_{G-V(C)}(x) \right\},$$

a finite union of bands, together with a flat band at each common zero of μ_G and $\mu_{G-V(C)}$, where $F(x, \cdot)$ vanishes identically.

Corollary 9. *For $b_1(G) = 1$ the roots of $\mu_{d,G}$ are exactly the solutions of $\mu_G(x) = 2 \cos(\frac{k\pi}{d+1}) \mu_{G-V(C)}(x)$ for $1 \leq k \leq d$. In particular every root of every $\mu_{d,G}$ lies in $\text{spec}(T)$, and as $d \rightarrow \infty$ the roots equidistribute on $\text{spec}(T)$ with respect to its density of states.*

Proof. Immediate from Theorem 6 in the form $\mu_{d,G} = \prod_k F(x, e^{ik\pi/(d+1)})$ and $F(x, e^{i\theta}) = \mu_G - 2 \cos \theta \mu_{G-V(C)}$. The equidistribution is the arcsine law for $\cos(k\pi/(d+1))$ transported through the fibre map. $\square$

Hall–Puder–Sawin place the roots of $\mu_{d,G}$ in the Ramanujan *interval* $[-\rho, \rho]$, ρ being the spectral radius of T . Corollary 9 places them in $\text{spec}(T)$, which is a union of bands and is a proper subset of that interval whenever T has a spectral gap: no root of any $\mu_{d,G}$ can ever enter a gap. The two statements coincide exactly when the band structure is trivial, and that is precisely the case of a cycle, where $\mu_{G-V(C)} = 1$ and $\text{spec}(T) = [-2, 2]$ is the whole

interval. Attaching even a single pendant edge opens gaps, and they are not small (computed band structures, proportion of $[-\rho, \rho]$ that is a gap):

G	ρ	gap proportion
C_n	2	0%
triangle + pendant	2.170	36.5%
C_4 + pendant	2.136	45.9%
triangle + P_2	2.214	51.3%
C_4 + P_3	2.189	54.1%
triangle + $K_{1,3}$	2.514	65.9%

So the cycle is not merely the easiest case of Theorem 6: it is the degenerate one, the unique member of the family whose spectrum has no gaps and for which the Ramanujan interval is attained. Any cycle with a tree attached exhibits a band structure that no cycle can display.

Related work. Two ingredients are classical and are used, not claimed. The factorization of the characteristic polynomial of a cyclic cover into twisted determinants at roots of unity is the block-circulant diagonalization [5, 6]; in the infinite-cover limit it is the Floquet–Bloch decomposition of a periodic discrete operator, and $y = -P/(2Q)$ is the corresponding Floquet discriminant. The identity $\Phi_{G,r} = \chi_G \cdot \mu_{r-1,G}$ is [2]. Closest in form is McSorley and Feinsilver [4]. For a path whose edges are cyclically labelled with t labels they establish a (τ, Δ) -recurrence $\Phi_N = \tau\Phi_{N-1} - \Delta\Phi_{N-2}$, which is the algebraic skeleton of (3), and their Theorem 4.4 solves it as $\Delta^{N/2}U_N(\tau/2\sqrt{\Delta})$, exactly the form used here. In the univariate specialization their τ is the matching polynomial of the cycle C_t and $\Delta = (-1)^t x_1 \cdots x_t$ is a monomial; passing between their counting normalization and the characteristic one used here is the homogenization $V_d \mapsto c^d V_d$, under which a monomial Δ becomes $\mu_{G-V(C)} = 1$. Their Theorems 4.4 and 4.5 together give $U_{dn+n-1} = U_d(T_n)U_{n-1}$, the identity invoked in Section 1 to pass between Theorem 1 and Hall’s conjecture, so that bridge was already available in 2009. Their parameter is the number of label periods along a path and their graphs are paths, cycles and rooted trees; there is no covering and no cycle with trees attached. What is added here is the pair $(\mu_G, \mu_{G-V(C)})$, whose second entry is a genuine polynomial recording the attached trees rather than a monomial, together with the covering interpretation that makes the recurrence a statement about $\mu_{d,G}$.

The boundary is sharp. For $b_1(G) = b$ the lift is determined by b permutations, that is by an action of the free group F_b on $[r]$, whose orbits are the components of the lift. The exponential formula still applies, in the form $N_r = \sum_{\ell} \binom{r-1}{\ell-1} C_{\ell} N_{r-\ell}$ with $N_r = (r!)^b \Phi_{G,r}$ and C_{ℓ} the sum of χ over the transitive b -tuples on $[\ell]$; we have verified this for $b \leq 3$. What fails is the atom. For $b = 1$ every transitive tuple gives the *same* cover G_{ℓ} , so $C_{\ell} = (\ell-1)! \chi_{G_{\ell}}$: one graph per ℓ , with $\chi_{G_{\ell}}$ Chebyshev in ℓ . For $b \geq 2$ the transitive tuples number $(\ell-1)!$ times the count of index- ℓ subgroups of F_b and spread over many non-isomorphic covers with distinct spectra, so C_{ℓ} carries no such structure. Accordingly Theorem 6 is not merely unproved for $b \geq 2$: it is false there already at $d = 1$, as we checked for the theta graph, the bowtie, K_4 , $K_{2,3}$ and two triangles joined by an edge.

Theorem 6 was verified by exact integer computation against the definition of $\mu_{d,G}$ as an average of matching polynomials over all d -covers, and against brute-forced lift averages, on

thirteen named unicyclic graphs for $r \leq 6$ and on a random sample of thirty-four for $r \leq 5$; Lemma 7 was checked on eleven.

4 The formalization

The core algebraic derivation — the exponential formula, the closed form of Theorem 1 for all n, r (equivalently, everything from the cycle weights (1) to the closed form), its factorization, and the parity Corollary 5 — is machine-checked in Lean 4 over Mathlib (v4.30). The main entry points:

- `Paper2.expFormula` — the exponential formula (2) in integer form $r A_r = \sum_k r^{(k)} f(k) A_{r-k}$ over any commutative ring, proved via Mathlib’s Cauchy cycle-type count (`Equiv.Perm.card_of_cycleType`);
- `Paper2.thm1_Sr` — Theorem 1 in the form: the expected cycle weight (1) equals the closed form, for all n, r , over any $\mathbb{Q}$ -algebra domain;
- `Paper2.cG_factored`, `Paper2.quotient_parity` — the factorization and Corollary 5.

The three remaining corollaries are formalized only in part: Corollary 2’s generating function appears as the finite recurrence (2) rather than a literal power-series identity; of Corollary 3 only the underlying trigonometric factorization (`cor1_factored`) is checked, not the “all roots in $[-2, 2]$ ” statement end-to-end; and Corollary 4 (`cor2_phi3`) is a bounded `native_decide` over $n, r \leq 6$.

Trust base for the general theorems: `propext`, `Classical.choice`, `Quot.sound` only (checked with `#print axioms`); the bounded `native_decide` checks additionally invoke `Lean.ofReduceBool`. No sorry. The single remaining unformalized ingredient of the *graph-theoretic* statement is the sentence before (1): a lift of a cycle is a disjoint union of cycles — elementary covering-space combinatorics for which Mathlib currently lacks the vocabulary.

Remark 10. Everything here is stated for the uniform permutation model; by [2] the quotient coincides with the $(r - 1)$ -matching polynomial, so Theorem 1 can alternatively be deduced from [3]. We believe the generating function of Corollary 2, the evaluation of Corollary 4, and the formalization are new.

References

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